

NETWORK VULNERABILITY ASSESSMENT AND SECURITY REPORT

Project Title	Network Vulnerability Assessment and Security Report
Student Name	[Your Full Name]
Date	[Submission Date]
Lab Environment	Kali Linux (Attacker) + Metasploitable 2 (Target: 192.168.0.7)
Tool Used	Nmap (Version Detection Scan)
Assessment Type	Authorized Educational Vulnerability Assessment

1. Introduction

Purpose of the Assessment

The purpose of this assessment is to identify open ports, running services, service versions, and potential security weaknesses on the target system (Metasploitable 2) in a controlled laboratory environment as part of a CEH training exercise.

Scope

The assessment was limited to the Metasploitable 2 virtual machine (IP: 192.168.0.7) running in an isolated lab network. No production or unauthorized systems were involved.

Authorization Statement

This assessment was conducted strictly for educational purposes as part of a Certified Ethical Hacker (CEH) assignment. All testing was performed on authorized systems (Metasploitable 2) within an isolated laboratory environment. No unauthorized systems were scanned or assessed.

2. Methodology

The assessment followed a structured approach consisting of host discovery, port scanning, service and version identification, vulnerability correlation based on identified versions, and risk rating. All activities were performed in an isolated lab environment using authorized tools.

3. Findings

3.1 Open Ports

The following ports were identified as open on the target system (192.168.0.7):

Port	Protocol	State	Notes
21	tcp	open	FTP
22	tcp	open	SSH
23	tcp	open	Telnet
25	tcp	open	SMTP
53	tcp	open	DNS
80	tcp	open	HTTP
111	tcp	open	RPCBind
139	tcp	open	NetBIOS / Samba
445	tcp	open	NetBIOS / Samba
512	tcp	open	Rexec
513	tcp	open	Login

514	tcp	open	Shell / tcpwrapped
1099	tcp	open	Java RMI
1524	tcp	open	Bindshell
2049	tcp	open	NFS
2121	tcp	open	FTP (ProFTPD)
3306	tcp	open	MySQL
5432	tcp	open	PostgreSQL
5900	tcp	open	VNC
6000	tcp	open	X11
6667	tcp	open	IRC
8009	tcp	open	AJP13
8180	tcp	open	HTTP (Tomcat)

3.2 Running Services

The following services were detected as running on the target:

Port	Service Name	Description
21	ftp	File Transfer Protocol
22	ssh	Secure Shell
23	telnet	Telnet remote access
25	smtp	Simple Mail Transfer Protocol
53	domain	DNS service
80	http	Web server
111	rpcbind	RPC port mapper
139 / 445	netbios-ssn	Samba file/print sharing
512	exec	Remote execution service
1099	java-rmi	Java Remote Method Invocation
1524	bindshell	Bind shell service
2049	nfs	Network File System
2121	ftp	ProFTPD FTP service
3306	mysql	MySQL database
5432	postgresql	PostgreSQL database
5900	vnc	Virtual Network Computing
6667	irc	Internet Relay Chat
8009	ajp13	Apache JServ Protocol
8180	http	Apache Tomcat web server

3.3 Service Versions

Service version information identified during the assessment:

Port	Service	Version Detected
21	ftp	vsftpd 2.3.4
22	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23	telnet	Linux telnetd
25	smtp	Postfix smtpd
53	domain	ISC BIND 9.4.2
80	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)

111	rpcbind	2 (RPC #100000)
139/445	netbios-ssn	Samba smbd 3.X – 4.X (workgroup: WORKGROUP)
512	exec	netkit-rsh rexecd
1099	java-rmi	GNU Classpath grmiregistry
1524	bindshell	Metasploitable root shell
2049	nfs	2–4 (RPC #100003)
2121	ftp	ProFTPD 1.3.1
3306	mysql	MySQL 5.0.51a-3ubuntu5
5432	postgresql	PostgreSQL DB 8.3.0 – 8.3.7
5900	vnc	VNC (protocol 3.3)
6000	X11	(access denied)
6667	irc	UnrealIRCd
8009	ajp13	Apache Jserv (Protocol v1.3)
8180	http	Apache Tomcat/Coyote JSP engine 1.1

3.4 Potential Vulnerabilities

Based on the identified services and versions, the following potential security concerns were noted (high-level only):

Finding ID	Description	Related Service/Port	Notes
VULN-001	Outdated FTP server version	21 / vsftpd 2.3.4	Known older version
VULN-002	Clear-text remote access service	23 / Telnet	Unencrypted protocol
VULN-003	Outdated web server	80 / Apache 2.2.8	Older Apache version
VULN-004	Legacy Samba version	139, 445 / Samba	Older Samba release
VULN-005	Unnecessary remote services	512, 513, 514	Legacy r-services
VULN-006	Exposed database services	3306 MySQL / 5432 PostgreSQL	Databases reachable
VULN-007	Multiple outdated services	Various ports	Several old software versions

3.5 Risk Level

Each finding was assigned a risk level based on potential impact and exposure:

Finding ID	Risk Level	Justification
VULN-001	High	Outdated FTP service commonly associated with known issues
VULN-002	High	Telnet transmits data in clear text
VULN-003	Medium	Older web server version increases exposure
VULN-004	Medium	Legacy Samba version may contain weaknesses
VULN-005	High	Unnecessary remote execution services increase attack surface
VULN-006	High	Database services exposed without apparent restriction
VULN-007	Medium	Multiple outdated services increase overall risk

Risk Rating Criteria:

- High – Significant potential impact or clear-text / unnecessary services.
- Medium – Moderate impact from outdated software versions.
- Low – Limited impact under normal conditions.

4. Risk Summary

The target system (Metasploitable 2) presents a high overall risk posture due to a large number of open ports, multiple outdated services, clear-text protocols, and exposed database services. This is expected for an intentionally vulnerable training system.

Summary of Findings:

High Risk	Medium Risk	Low Risk
4	3	0

5. Recommended Remediation

The following high-level remediation actions are recommended:

Finding ID	Recommended Action	Priority
VULN-001	Update or replace the FTP service with a current, supported version	High
VULN-002	Disable Telnet and use SSH for remote access instead	High
VULN-003	Upgrade Apache to a supported current version and apply security patches	Medium
VULN-004	Update Samba to a current supported release and restrict access	Medium
VULN-005	Disable unnecessary r-services (rexec, rlogin, etc.)	High
VULN-006	Restrict database access to trusted hosts only; enforce strong authentication	High
VULN-007	Perform a full system update and remove unused services	Medium

General Recommendations:

- Apply security patches and updates for all identified services.
- Disable or restrict unnecessary services and ports.
- Implement strong authentication and access controls.
- Configure host-based firewalls to limit exposure.
- Follow the principle of least privilege for all services and accounts.

6. Conclusion

The network vulnerability assessment of the Metasploitable 2 system identified a significant number of open ports and outdated services, resulting in a high overall risk rating. This outcome is consistent with the system’s design as an intentionally vulnerable training platform. Regular vulnerability assessments remain essential for identifying and addressing security weaknesses. All testing was conducted ethically and strictly within the authorized educational lab environment.

7. Appendix

A. Screenshots of Scan Results

```
specter@kali: ~  
Session Actions Edit View Help  
$ nmap -sV 192.168.0.7  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-14 06:55 -0400  
Nmap scan report for 192.168.0.7  
Host is up (0.00025s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshcd  
513/tcp   open  login          
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath gmicregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied) and already know the username is admin, use this  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 08:00:27:3B:AA:40 (Oracle VirtualBox virtual NIC)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.87 seconds  
$
```

B. Additional Notes

Target hostname: metasploitable.localdomain | MAC: 08:00:27:3B:AA:40 (VirtualBox)

C. References

Nmap documentation – https://nmap.org